Econ 2 - Lecture 12 - 2/23/26

Lecture Quiz # 7 Released Today

↳ Due Monday, March 2^{nd}

Last Class: Bad tech + Fiscal Policy

Today: Deficit & Debt = Chapter 5.1

Expenditure Multiplier = $\frac{1}{1-mpc}$

Tax Multiplier = $\frac{-mpc}{1-mpc}$

Fiscal Policy ⇒ Difficult to implement

Great Financial Crisis → Crisis ⇒ $\Delta Y = -1$ Trillion

Action

ARRA / Stimulus Bill

$\Delta G = 500$ Bil, $\Delta T = -300$ Bil

Low mpc = ? , mpc = 0.5

Expenditure Multiplier = $\frac{1}{1-0.5} = 2$

Tax Multiplier = $\frac{-0.5}{1-0.5} = -1$

$$\left.\begin{array}{l}\Delta Y = 2 \times 500 \\ \Delta Y = -1 \times (-300)\end{array}\right\} = 1000 + 300 = 1{,}300\text{B} = 1.3\text{T}$$

COVID fiscal Policy

March/April 2020 = \$2 Trillion
December 2020 = \$1 Trillion
March 2021 = \$2 Trillion
= \$5 Trillion

Consequence → Government needs to borrow!

Deficit = G − T > 0

All gov't spending
→ transfer payments
→ Interest on Debt

Tax Revenue

2020: G > T ⇒ 3 Trillion Deficit
2021: G > T ⇒ 2.77 Trillion
22: G > T ⇒ 1.4 T
23: G > T ⇒ 1.7 T

Normalize Deficit → Divide by GDP

Today $\frac{\text{Deficit}}{\text{GDP}} \sim 6\%$ → only high (> 3%) during recessions + wars

↓
Low UE!

Debt = sum of deficits = $\sum (G - T)$

3 Yr Example

$$\text{Debt} = \underbrace{(G_1 - T_1)}_{=50} + \underbrace{(G_2 - T_2)}_{=100} + \overbrace{\underbrace{(G_3 - T_3)}_{=-75}}^{\text{Surplus}}$$

Borrow \$50 in Yr 1

Borrow \$100 in Yr 2

Pay back \$75 in Yr 3

Debt = 50 + 100 − 75 = \$75

Stock Variable

Flow Variable

Facts about National Debt

Current Debt = \$38.7 T

Fact #1: US Government <u>DOES NOT</u> owe \$38.7 to lenders

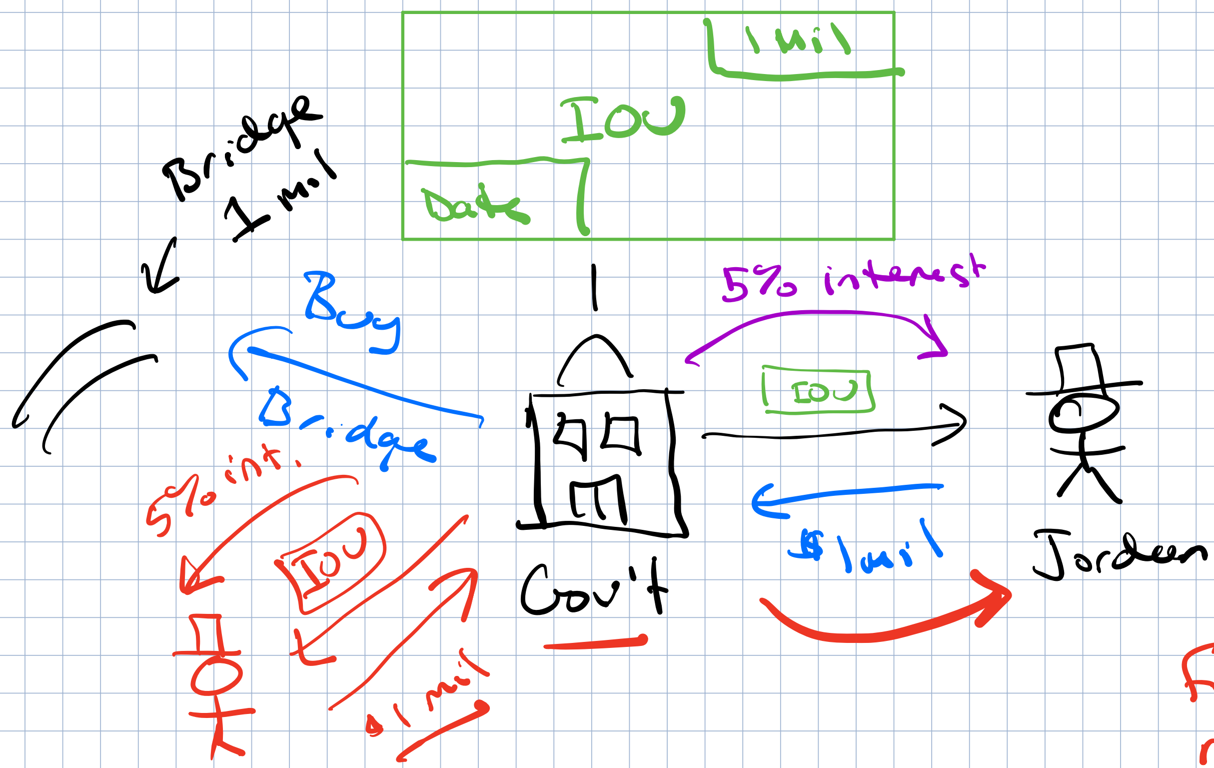
$$\text{Total Debt} = \underbrace{\text{Public Debt}}_{\substack{\text{Borrowing from outside the Gov't (Bonds)} \\ \$31.1\text{T}}} + \underbrace{\text{Intra-Governmental Borrowing}}_{\substack{\text{Borrowing from within the Gov't} \\ \to \text{Social Security Trust, etc.} \\ = \$7.6\text{T}}}$$

Fact #2: We **DO NOT** need to pay back the $31 T in public debt?

Setting: Government borrow $1 million from lenders

Government Bond: vehicle for gov't to borrow money

"I owe you" (IOU) → 1 mil in one year



In 2027, Jordan wants $1 mil back

Find a new lender!

Roll over the debt indefinitely

→ Concerns about paying interest
→ Rolling debt over
} Hard to find Lenders

As long as we believe they will be solvent

What do we need to worry about?

Interest Payments on debt

Where does the gov't generate money to pay interest? Tax Revenue!

Public Debt = \$1 million
Interest Rate = 5% = 0.05
Nominal GDP = \$5 million (Income today)

Interest Payment on Debt = Interest Rate × Debt

= 0.05 × 1 mil = \$50,000

As long as gov't can pay \$50,000 they can hold a debt of \$1 mil

→ Need to tax citizens → generate \$50,000

$$\underbrace{\text{Tax Revenue}} \geq \text{Interest Payment}$$

Tax Rate × Nominal GDP = Interest Payment

$$\text{Minimum Tax Rate} = \frac{\text{Interest Payment}}{\text{Nominal GDP}} \times 100$$

↓

Min. Tax Rate must be 1% in order to hold \$1 mil in debt

$$= \frac{50{,}000}{5{,}000{,}000} \times 100 = 1\%$$

	Yr 1	Yr 2
Nom. GDP	200,000	300,000
Public Debt	100,000	200,000
Interest Rate	5%	5%
Price Level	100	120

What is the minimum tax rate?
Tax burden from interest on debt?

$$\text{Yr 1 Min. Tax Rate} = \frac{\text{Int. Rate} \times \text{Debt}}{\text{Nom. GDP}} \times 100$$

$$= \frac{5\% \times 100{,}000}{200{,}000} \times 100$$

$$= 2.5\%$$

$$\text{Yr 2 Min. Tax Rate} = \frac{5\% \times 200{,}000}{300{,}000} \times 100$$

$$= \frac{10{,}000}{300{,}000} \times 100 =$$

$$= 3.33\%$$

Debt burden increased from 2.5% to 3.33%

What is the Real GDP in year 2? (Compare to Yr 1)

	Yr 1	Yr 2
Nom. GDP	200,000	300,000
Public Debt	100,000	200,000
Interest Rate	5%	5%
Price Level (GDP Deflator)	100	120

$$\text{Price Level} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

$$\text{Real GDP} = \frac{\text{Nom. GDP}}{\text{Price Level}} \times 100$$

$$\text{Real GDP}_2 = \frac{300{,}000}{120} \times 100$$

$$\text{Real GDP}_2 = 250{,}000$$

$$\text{Real GDP}_1 = 200{,}000$$

In yr 1: Real GDP = 200,000, Tax = 2.5%

In yr 2: Real GDP = 250,000, Tax = 3.33%

Was the tax increase worth it?

Fact # 3: $\frac{\text{Debt}}{\text{GDP}}$ ratio $\Rightarrow$ Appropriate measure of Debt <u>Burden</u>